Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/09/2004
PI Name:	Wang, Ning Ph.D.		
Project Title:	Control of Cell Mechanics and Function by Cell Microenvironment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	nwang@hsph.harvard.edu	**Fax:**	FY 617-432-3468
PI Organization Type:	UNIVERSITY	**Phone:**	617-432-1686
Organization Name:	Harvard Univerity		
PI Address 1:	School of Public Health, Physiology Program		
PI Address 2:	665 Huntington		
PI Web Page:	http://www.hsph.harvard.edu/		
City:	Boston	**State:**	MA
Zip Code:	02115	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	08/01/2001	**End Date:**	07/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Butler, James (Harvard University)		
Grant/Contract No.:	NAG 2-1509		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The tasks of this project are: 1) to identify specific transmembrane molecules that transduce mechanical stress into alterations in endothelin-1 gene expression; 2) to determine to what degree endothelin-1 gene expression depends on the stiffness of the extracellular matrix; 3) to determine to what degree cytoskeletal tension and stiffness depends on the stiffness of the extracellular matrix and external mechanical stimulation.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The objective of this project is to elucidate how cell microenvironment controls and regulates cell functions. Elucidating the molecular pathway by which mechanical forces regulate gene expression and determining control of gene expression, cytoskeletal tension, and stiffness by the extracellular matrix mechanics should help us understand the molecular basis of mechanotransduction and mechanoregulation in microgravity as well as in phyiological and pathological conditions.
Task Progress:	During the past funding period, we have developed a novel 3-dimensional magnetic twisting device for generating mechanical torques in any direction. A ferrimagnetic bead coated with a peptide containing the sequence Arg-Gly-Asp (a ligand for transmembrane molecule integrins) was bound to the apical surface of the cell. Using three pairs of orthogonally-aligned coils, an oscillatory mechanical torque was applied to a bead about any chosen axis at frequencies up to 1 kHz. The resulting oscillatory deformations of the cell were measured. In both unpatterned and micropatterned elongated cells on extracellular matrix, the mechanical stiffness perpendicular to the long axis of the cell was roughly two times smaller than that parallel to the long axis. Elongated cells on poly-L-lysine lost stress fibers and exhibited little mechanical anisotropy. Disrupting the actin cytoskeleton or decreasing cytoskeletal tension by overexpressing caldesmon also decreased the anisotropy. These data suggest that the preferential orientation of stress fibers along the long axis of the cell, and the presence of intrinsic cytoskeletal tension along these fibers, may be the primary origins of mechanical anisotropy in adherent living cells. (Hu et al., submitted). We also found that the extracellular matrix of the cell microenvironment and the intracellular microtubules play complementary roles in balancing the prestress, the pre-existing cytoskeletal tension. The controversy surrounds the cellular tensegrity model. Some argue that microtubules (MTs) must bear a significant portion of cell contractile stress (prestress) if tensegrity is a useful model. Previously we have shown that for highly spread airway smooth muscle cells (areas>2500 mm2) MTs balance a significant but small potion (average 14%) of the prestress. To further explore if controlling the degree of cell spreading could modulate the portion of the prestress balanced by MTs, we utilized a recent method by which tractions are quantified in cells that are constrained within micropatterned adhesive islands of defined sizes on the surface of flexible polyacrylamide gels containing fluorescent microbeads. The prediction is that if MTs balance a portion of the contractile stress, then, upon their disruption, the portion of the stress balanced by MTs would shift to the substrate, causing an increase in traction and strain energy. We first activated the cells maximally with histamine and then disrupted the MTs with colchicine. Histamine resulted in an increase in intracellular calcium whereas colchicine addition in the presence of histamine did not change intracellular calcium concentration, suggesting there was no additional net increase in contractile stress inside the cell. We found that following disruption of MTs the increase in traction and strain energy varied with the degree of cell spreading: as the cell projected areas increased from 500 mm2 to about 1800 mm2, the percent increase in tractions decreased from 80% to about a few percent and the percent increase in strain energy decreased from 200% to almost zero percent, indicating the portion of the prestress balanced by MTs decreased as the cells increased spreading. These findings demonstrate that complementary role of the extracellular matrix and the MTs in balancing the prestress is controlled by the degree of cell spreading. (Hu et al., Front. Biosci., 2004).
Bibliography Type:	Description: (Last Updated: 02/25/2021)
Abstracts for Journals and Proceedings	Stamenovic, D.; Suki, B.; Fabry, B.; Buy, J.E.; Wang, N.; Fredberg, J.J. "The role of contractile stress in cell transition from solid-like to fluid-like behavior." Proceedings of 2003 BMES Annual Fall Meeting (CD-ROM). 2003 BMES Annual Fall Meeting Nashville, TN. , Oct-2003
Articles in Peer-reviewed Journals	Hu, S.; Chen, J.; Fabry, B.; Numaguchi, Y.; Gouldstone, A.; Ingber, D.E.; Fredberg, J.J.; Butler, J.P.; Wang, N. "Intracellular stress tomography reveals stress focusing and structural anisotropy in the cytoskeleton of living cells." Am J Physiol Cell Physiol 285(5):C1082-C1090, Nov 2003. , Nov-2003
Articles in Peer-reviewed Journals	Tolic-Norrelykke, I.M.; Wang, N. "Traction, Trypsin, and Tensegrity." Internet Electron J Mol Des 2(10):642-652, Oct 2003. , Oct-2003
Articles in Peer-reviewed Journals	Hu, S.; Chen, J.; Wang. N. "Cell spreading controls balance of prestress by microtubules and extracellular matrix." Front Biosci 9:2177-82, Sep 2004. , Sep-2004
Articles in Peer-reviewed Journals	Tolic-Norrelykke, I.M.; Wang, N. "Traction in smooth muscle cells varies with cell spreading." J Biomech, in press. , Dec-2004
Presentation	Wang, N. "Cytokeletal prestress determines cell shape stability and stress distribution." University of Illinois at Champaign. Sep-2003
Presentation	Wang, N. "Central Role of Cytoskeletal Stress in Cell Functions." The 50th anniversary of Huazhong University of Science and Technology. Oct-2003
Presentation	Wang, N. "Intracellular stress tomography reveals stress focusing in the cytoskeleton of living cells." INSERM, France. Jan-2004